

B.Sc. (Hons.) (Semester-V) Examination, 2020-21**Physics****Paper : BPT-501
Mathematical Physics****Time : Four Hours****Full Marks : 70****Instructions :**

1. The Question paper contains **08** questions out of which you are required to answer **any 04** questions. The question paper is of **70** marks with each question carrying **17.5** marks.
2. The total duration of the examination will be **4 hours** (Four hours), which includes the time for downloading the question paper from the Portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
3. For the students with benchmark disability as per Persons with Disability Act, the total duration of examination shall be **6 hours** (six hours) to complete the examination process, which includes the time for downloading the question paper from the Portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
4. Answer should be hand-written on a plain white **A4** size paper using **black** or **blue** pen. Each question can be answered in upto **350** words on **3** (Three) plain **A4** size paper (**only one side is to be used**).
5. Answer to each question should start from a fresh page. All pages are required to be numbered. You should write your Course Name, Semester, Examination Roll Number, Paper Code, Paper title, Date and Time of Examination of the first sheet used for answers.

Questions :**Q.1.** 1. Show that for a scalar function ψ

$$(\vec{r} \times \vec{\nabla}) \cdot (\vec{r} \times \vec{\nabla})\psi = r^2 \nabla^2 \psi - r^2 \frac{\partial^2 \psi}{\partial r^2} - 2r \frac{\partial \psi}{\partial r}$$

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2. Find the Newton's equations of motion for a particle of mass m moving under the influence of a potential V . (When written in Spherical polar coordinates they relate the partial derivatives of V , with respect to (r, θ, ϕ) , with the corresponding components of the force on the particle).

12.5**Q.2.** 1. The Components of vectors **A** and **B** and a second rank tensor T_{mn} are given in one coordinate system by

$$A = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, B = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } T = \begin{pmatrix} 2 & \sqrt{3} & 0 \\ \sqrt{3} & 4 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

in a second coordinate system obtained by proper rotation of the first coordinate system, the components of **A** and **B** are

$$A' = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \\ 0 \end{pmatrix}, B' = \begin{pmatrix} -\sqrt{3}/2 \\ 1/2 \\ 0 \end{pmatrix}$$

(i) Find the rotation matrix.

(ii) Find the components of the second rank tensor T'_{ij} in the new coordinate system.

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2. Two quantities A and B transform according to the transformation equations given below in (i) and (ii) under coordinate transformations from x^i to $\bar{x}^j$. Express A and B in the index notation, by identifying the contravariant and covariant indices, and hence the rank:

(i) $A'(\bar{p}, \bar{q}, \bar{r}) = \frac{\partial x^k \partial x^l \partial \bar{x}^r}{\partial \bar{x}^p \partial \bar{x}^q \partial x^m} A(k, l, m)$ **3**

(ii) $B'(\bar{p}, \bar{q}, \bar{r}, \bar{s}) = \frac{\partial x^k \partial \bar{x}^p \partial \bar{x}^q \partial \bar{x}^s}{\partial \bar{x}^r \partial x^l \partial x^m \partial x^n} B(k, l, m, n)$

3. Find the number of independent components of a totally antisymmetric (with respect to each pair of indices) tensor of rank 3 in d dimensions. **2.5**

P.T.O.

Q.3. Starting from the vector identity

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{V}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{V}) - \nabla^2 \vec{V},$$

find an expression for $\nabla^2 \vec{V}(r, \theta, \phi)$ in spherical polar coordinates. **17.5**

- Q.4.** 1. Find the eigenvalues and eigenvectors of the matrix, $A = \begin{pmatrix} 7 & \sqrt{6} & -\sqrt{3} \\ \sqrt{6} & 2 & -5\sqrt{2} \\ -\sqrt{3} & -5\sqrt{2} & -3 \end{pmatrix}$,
and write down the diagonal form of the matrix A . **12.5**

2. Show that in Cartesian coordinates, under coordinate transformations from x^i to $\bar{x}^j$,

$$\frac{\partial x^m}{\partial \bar{x}^i} \frac{\partial x^m}{\partial \bar{x}^j} = \delta_{ij}, \quad (i, j, m = 1, 2, 3)$$

What does this imply? What is the metric tensor in Cartesian coordinates? **5**

P.T.O.

- Q.5.** 1. Solve $(1-x^2)y'' - 2xy' + p(p+1)y = 0$, with p an integer, about the point $x = 1$, using the Frobenius method. Write down the solutions of the above differential equation for $p = 1, 2, 3$. **12.5**

2. Find a linearly independent second solution of the above differential equation for $p = 1$. 5
- Q.6.** 1. Find the contravariant components of $F = \frac{-y}{x^2+y^2}\hat{a}_x + \frac{x}{x^2+y^2}\hat{a}_y$ in circular cylindrical coordinates. By the operation of lowering of indices find the corresponding covariant components. 9.5
2. Make a neat sketch of the function $f(x) = \cos^2(x)$, and find its Fourier Series. 8
- Q.7.** 1. Find the nature of the singularity at the point at infinity for the following differential equations (in which p is an integer),
- (i) $(1 - x^2)y'' - 2xy' + p(p + 1)y = 0$,
- (ii) $x^2y'' + xy' + (x^2 - p^2)y = 0$. 9.5
2. Using the generating function $g(x, t) = \sum_{n=0}^{\infty} P_n(x)t^n$, ($|t| < 1$, $|x| < 1$), find the orthogonality condition for Legendre Polynomials. 8
- Q.8.** 1. Using the generating function $g(x, t) = e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!}$ for the Hermite polynomials, evaluate $\int_{-\infty}^{\infty} xe^{-x^2}H_n(x)H_m(x)dx$. 8
2. Find the Fourier series of $f(x) = x^2$, $(-\pi < x < \pi)$ and using the series evaluate $\sum_{n=1}^{\infty} \frac{1}{n^2}$. 9.5

External Portals & Academic Resources:

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app>